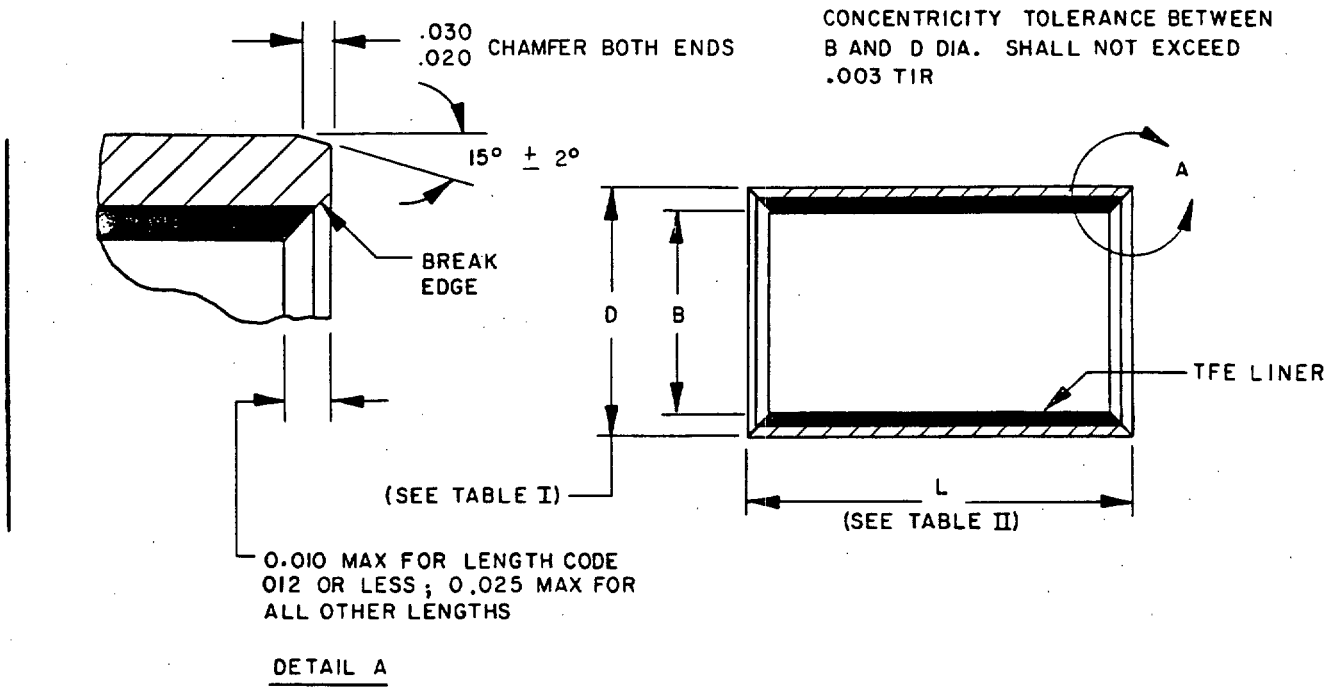
MILITARY SPECIFICATION SHEET

BEARINGS, SLEEVE, PLAIN, SELF-LUBRICATING, 325°F

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of MIL-B-81934.



DETAIL A

AMSC N/A

FSC 3120

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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TABLE I. Sleeve bearing dimensions.

Dash no.	Nominal size	B (Bore diameter) +.0000 -.0010	D dia 1/	Weight lb/in (ref) L = 1.000	
				Al	Cres
-04	(1/4)	.2515	.3760	.006	.016
-05	(5/16)	.3140	.4386	.007	.019
-06	(3/8)	.3765	.5012	.008	.022
-07	(7/16)	.4390	.5638	.009	.025
-08	(1/2)	.5015	.6265	.011	.028
-09	(9/16)	.5640	.6892	.012	.031
-10	(5/8)	.6265	.8142	.021	.056
-11	(11/16)	.6890	.8767	.022	.060
-12	(3/4)	.7515	.9393	.024	.065
-14	(7/8)	.8765	1.0645	.028	.075
-16	1	1.0015	1.1898	.031	.084
-18	(1-1/8)	1.1265	1.3148	.035	.094
-20	(1-1/4)	1.2515	1.4398	.038	.103
-22	(1-3/8)	1.3765	1.5648	.041	.113
-24	(1-1/2)	1.5015	1.7523	.062	.171
-26	(1-5/8)	1.6265	1.8773	.067	.183
-28	(1-3/4)	1.7515	2.0023	.071	.196
-32	2	2.0015	2.2523	.081	.222

1/ D tolerance: Aluminum ±.0005; Corrosion resistant steel +.0000, -.0005.

Example of weight calculation: M81934/1-16A008

Sleeve weight: (0.031 lb/in) x .250 inch = 0.008 lb

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TABLE II. Sleeve bearing length.

Dash no.	Nominal size	Length L +.000 -.010														
		.156	.187	.218	.250	.281	.312	.343	.375	.437	.500	.562	.625	.687	.750	.875
-04	1/4	005	006	007	008	009	010	011	012	014						
-05	5/16	005	006	007	008	009	010	011	012	014	016	018				
-06	3/8	005	006	007	008	009	010	011	012	014	016	018	020	022		
-07	7/16	005	006	007	008	009	010	011	012	014	016	018	020	022	024	028
-08	1/2	005	006	007	008	009	010	011	012	014	016	018	020	022	024	028
-09	9/16	005	006	007	008	009	010	011	012	014	016	018	020	022	024	028
-10	5/8	005	006	007	008	009	010	011	012	014	016	018	020	022	024	028
-11	11/16				008	009	010	011	012	014	016	018	020	022	024	028
-12	3/4				008	009	010	011	012	014	016	018	020	022	024	028
-14	7/8				008	009	010	011	012	014	016	018	020	022	024	028
-16	1				008	009	010	011	012	014	016	018	020	022	024	028
-18	1 1/8						010	011	012	014	016	018	020	022	024	028
-20	1 1/4								012	014	016	018	020	022	024	028
-22	1 3/8								012	014	016	018	020	022	024	028
-24	1 1/2								012	014	016	018	020	022	024	028
-26	1 5/8										016	018	020	022	024	028
-28	1 3/4										016	018	020	022	024	028
-32	2										016	018	020	022	024	028

TABLE II. Sleeve bearing length (continued).

Dash no.	Normal size	Length L +.000 -.010														
		1.000	1.125	1.250	1.375	1.500	1.625	1.750	1.875	2.000	2.125	2.250	2.375	2.500	2.750	3.000
-04	1/4															
-05	5/16															
-06	3/8															
-07	7/16															
-08	1/2															
-09	9/16	032	036													
-10	5/8	032	036	040	044											
-11	11/16	032	036	040	044	048	052									
-12	3/4	032	036	040	044	048	052									
-14	7/8	032	036	040	044	048	052									
-16	1	032	036	040	044	048	052	056	060							
-18	1 1/8	032	036	040	044	048	052	056	060							
-20	1 1/4	032	036	040	044	048	052	056	060	064	068					
-22	1 3/8	032	036	040	044	048	052	056	060	064	068					
-24	1 1/2	032	036	040	044	048	052	056	060	064	068	072	076	080	088	
-26	1 5/8	032	036	040	044	048	052	056	060	064	068	072	076	080	088	096
-28	1 3/4	032	036	040	044	048	052	056	060	064	068	072	076	080	088	096
-32	2	032	036	040	044	048	052	056	060	064	068	072	076	080	088	096

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REQUIREMENTS

Material:

Bearing:

A = Aluminum alloy QQ-A-200/3 or QQ-A-225/6 (2024T851 or 2024T8511).

NOTE: The manufacturer may substitute Alloy 2024T351 or 2024T3511 for 2024T851 or 2024T8511, respectively, provided the processing procedures employed in manufacturing the bearing result in a conversion of the alloy to 2024T851 or 2024T8511.

C = Corrosion resistant steel, AMS 5643 (17-4PH) Condition H-1150 per MIL-H-6875.

Liner: See procurement specification.

Finish: Aluminum to be anodized per MIL-A-8625, Type I or II; or chemical film treatment per MIL-C-5541.

Surface texture: Smooth machine finish 63 micro-inch R_a on OD; 125 micro-inch R_a on all other metallic surfaces per ANSI B46.1, unless otherwise specified.

Temperature range: -65°F to +325°F.

Break sharp edges and corners and remove all burrs and slivers.

Dimensions in inches, unless otherwise specified.

Example of part number:

M81934/1 - 04 A 008 = 1/4 inch nominal size aluminum by 1/4 long.

- 008 — Length code in multiples of 1/32 inch
- A — Material code
- 04 — Nominal size code in multiples of 1/16 inch
- M81934/1 — Basic part number

Marking: For dash number 08 and larger the military part number shall be marked on the part.

Interchangeability relationship: Use only assigned part numbers for length as listed in length column of Table II.

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Load ratings: Qualification test dynamic loads are based upon the formula:

Test load (lbs) = 37,500 psi x bore diameter x (L - .1).

Use of these bearings will usually require operation at a lower stress level in order to obtain the desired wear/life characteristics for a specific application.

Static limit load: Aluminum = 50,000 B x (L-.1) lbs.
CRES = 78,500 X bore diameter x (L-.01) lbs.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activities:
Army - AR, AT
Navy - SH
Air Force - 99, 84
DLA - IS

Preparing activity:
Navy - AS
(Project 3120-0647)

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